

HANEEN NAJJAR

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SUMMARY

PhD student in Computer Science at Tel Aviv University and trustworthy ML engineer, working on adversarial robustness and fairness, AI safety, and security across computer vision, LLMs, and multi-agent systems. Author of Sy-FAR (USENIX Security 2026), a symmetry-based approach for fair adversarial robustness in computer vision.

Over eight years of industry experience at Nvidia and Hailo, leading end-to-end solutions in chip performance optimization, large-scale model deployment, and AI model optimization for edge devices. Active mentor for students and for women entering the tech industry through the Alpha undergraduate program and Alice Code.

EDUCATION

Ph.D., Computer Science & AI October 2025 – Present
Tel Aviv University, Tel Aviv, Israel

M.Sc., Computer Science 2023 – 2025
Tel Aviv University, Tel Aviv, Israel GPA 95

Thesis: Sy-FAR: Symmetry-based Fair Adversarial Robustness (accepted to USENIX Security 2026)

MBA, Management of Technology, Innovation, and Entrepreneurship 2020 - 2021
Tel Aviv University, Tel Aviv, Israel

B.Sc., Computer Science and Mathematics 2013 – 2017
University of Haifa, Haifa, Israel

PUBLICATIONS

Sy-FAR: Symmetry-based Fair Adversarial Robustness 2025
H. Najjar, E. Ronen, M. Sharif — *Accepted to USENIX Security 2026*

- Introduces a symmetry-based framework ensuring adversarial misclassifications between any two classes are balanced, improving fairness across demographic subgroups without sacrificing robustness.
- Evaluated across five datasets and three architectures under targeted and untargeted attacks; achieves significant gains in fair adversarial robustness with lower computational cost than state-of-the-art methods.

PROFESSIONAL EXPERIENCE

Max Planck Institute for Intelligent Systems, COMPASS Research Group: Research Intern 2026
• Conducted research on trustworthy machine learning for LLMs and multi-agent systems, focusing on safety, security, and robustness.

Hailo Technologies, Tel Aviv, Israel: Applied Machine Learning Engineer – Pretrained Models & Optimization
Jan 2023 – Oct 2025

- Led development of the **Hailo Model Zoo**, an open-source platform of pre-trained and optimized deep learning models (classification, detection, segmentation) for Hailo edge AI devices.
- Built end-to-end pipelines for **model quantization, optimization, profiling, and compilation** into Hailo Executable Format (HEF) binaries for real-time edge inference.
- Developed retraining workflows for fine-tuning on custom datasets and benchmarking workflows measuring accuracy at full precision, quantized, and on-device levels.

Nvidia, Tel Aviv, Israel: Software Team Lead (Jul 2019 – Jan 2023)

Software Engineer (Oct 2017 – Jun 2019) Oct 2017 – Jan 2023
• Led a software team for switch chip features end-to-end (research, design, development, testing) using Agile Scrum.

- Optimized chip performance algorithms (dynamic power management, FPS, latency), **improving throughput by 35%** and reducing power consumption.
- Built scalable Python/C/C++ automation tools, **improving engineering efficiency by 40%**; designed QoS net-working features for critical data prioritization.

TECHNICAL SKILLS

Programming: Python, C, C++, Java, MATLAB, Bash

Machine Learning: Deep Learning, Computer Vision, Adversarial ML, Trustworthy ML (Robustness, Fairness, Security), GNNs, NLP

Frameworks & Libraries: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas

Model Optimization & Deployment: Quantization, Model Compression, Edge AI Deployment, Performance Profiling & Benchmarking

MLOps & Tools: Git, CI/CD (Jenkins), Reproducible ML Workflows, AWS S3, Linux

HONORS, AWARDS & CERTIFICATES

Neubauer Fellowship	Aug 2026
Neubauer Family Foundation — awarded in support of PhD studies and research	
Deep Learning Model for Edge AI (Vision & Automotive) – Hackathon Award	2025
Hailo Technologies	
Software Hackathon Certificate	2020
Nvidia	
National Mathematical Olympiad Finalist	2019
Herzliya, Israel	

VOLUNTEERING

Research Mentor — Alpha Undergraduate Mentorship Program	Oct 2025 – Present
Mentor undergraduate students in research, advising on research methods, project development, and academic growth	
Education Mentor — Alice Code	Mar 2023 – 2026
Leadership and technology program for girls and young women; technical coaching and career guidance	